

Hydropower Site Suitability

Alin-Ioan Alexandru
Stefan-Cristian Grecu

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1 Introduction

This document describes the methodology, data processing pipeline, model training, and evaluation for the project focused on predicting site suitability for new hydropower plants using satellite imagery and deep learning segmentation models.

The objective is to identify optimal locations for new hydropower plants by analyzing a variety of environmental factors using supervised learning. While there is limited available research that directly uses machine learning models in the analysis of site suitability for hydropower sites, our method, with the use of satellite imagery, has the advantage that it can potentially uncover suitable locations that may not be considered by typical manual or rule-based analyses.

We decided to use two approaches, resulting in two ML pipelines. The two pipelines start the same by making requests to the Sentinel Hub API to acquire the satellite data. The first approach models the gathered data in a tabular format and uses traditional ML algorithms (Random Forest, XGBoost) to make predictions. The second approach models the data as images with multiple channels and uses a deep learning segmentation model (U-Net) to make predictions.

2 Data Sources and Gathering

The resulting dataset consists of about 3000 positive samples consisting of locations of hydropower plants from Europe and 5000 negative samples of random locations from European rivers.

2.1 Satellite Data

Using the Sentinel Hub API, we gather Sentinel-2 satellite imagery for the regions of interest. We focus on bands B3, B4, B8 and B11, which cover the visible and near-infrared spectrum.

The following spectral indices are calculated:

- NDVI (Normalized Difference Vegetation Index) - used to assess vegetation health and density. Vegetation cover can be used to assess land suitability for hydropower plants.
- NDWI (Normalized Difference Water Index) - used to identify water bodies and assess their extent. Water detection is important in identifying potential sites for hydropower plants.
- NDBI (Normalized Difference Built-up Index) - used to detect built-up areas and urbanization. This helps in avoiding locations that are too close to urban areas.
- MNDWI (Modified Normalized Difference Water Index) - used to enhance water features in urban areas.

2.2 Topographic Data

Topographic data is obtained from the Copernicus Digital Elevation Model (DEM). The DEM provides elevation data which is crucial for assessing the potential energy generation capacity of a site and the potential of forming and accumulation lakes near the plants. From the DEM, we also derive slope data, which helps in understanding the terrain and its suitability for hydropower infrastructure.

2.3 Hydrological and Precipitation Data

Hydrological data, specifically river discharge data, is obtained from the Hydorives database and precipitation data is gathered using the Open-Meteo API. River discharge is a key factor in determining the potential energy output of a hydropower plant, while precipitation data helps in understanding the water availability in the region.

2.4 Labels

Positive samples: near existing hydropower plants.

Negative samples: random locations at least 30 kilometers away from any hydropower plant.

Mask values (for the segmentation) and the classes used are as following:

1 = suitable, 0 = unsuitable.

3 Data Preprocessing

3.1 Preprocessing for Tabular Data

Preprocessing steps:

- Splitting data: Performed train-test split (e.g., 80-20).
- Handling missing values: Imputed missing values using mean strategy.
- Outlier detection: Identified and treated outliers using IQR method and replaced them using the same imputation strategy as for the missing values.
- Normalization: Scaled features to a standard range.

3.2 Preprocessing for Image Data

Feature cube construction: shape (H, W, C) with channels for spectral indices, DEM, slope, discharge, precipitation.

4 Machine Learning Models

4.1 Tabular Data Models

For the tabular representation of the dataset, we trained and compared three widely used ensemble learning algorithms:

- **Random Forest:** An ensemble of decision trees using bootstrap sampling and feature bagging. It provides robust performance, reduces overfitting, and allows for feature importance ranking.
- **MLP (Multi-Layer Perceptron):** A feedforward neural network with multiple layers and non-linear activation functions. It can model complex relationships in the data and is suitable for tabular data.
- **XGBoost:** An optimized implementation of gradient boosting decision trees. It generally provides higher accuracy, faster convergence, and better handling of imbalanced data.

4.2 Segmentation Model

For the image-based approach, we used a **U-Net** architecture implemented via the Segmentation Models PyTorch (SMP) library. The encoder was initialized with weights pre-trained on ImageNet to accelerate convergence.

- **Input:** Multi-channel feature cubes composed of spectral indices, DEM, slope, discharge, and precipitation layers.
- **Loss function:** Combination of Cross-Entropy Loss and Dice Loss to balance pixel-wise classification and overlap quality.
- **Optimizer:** Adam optimizer with an initial learning rate of 1×10^{-3} and learning rate scheduling.
- **Training:** Conducted for 20–30 epochs with a batch size of 16.

The model outputs a binary mask where each pixel is classified as either suitable or unsuitable for hydropower site placement.

5 Evaluation and Results

5.1 Tabular Data Models

5.2 segmentation Model

6 Conclusion